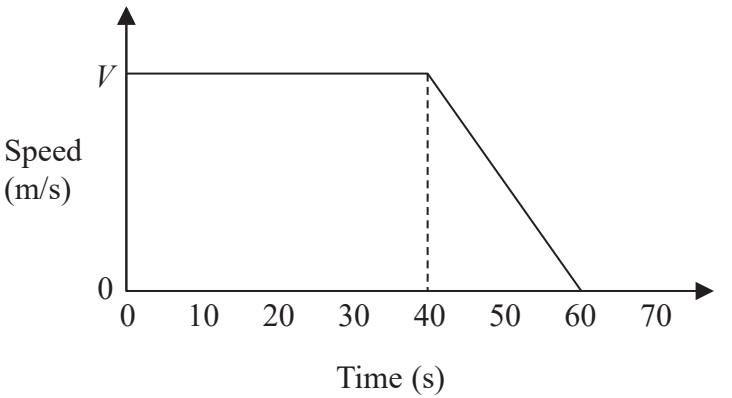
18 Here is a speed-time graph for part of a car journey.
This part of the journey took 60 seconds.



The car travelled at a constant speed of V m/s for the first 40 seconds.
It travelled 1 km in the 60 seconds.

Work out the value of V.

Distance = area under the speed–time graph.

1 mark
Area represents 1000 m.

$40V + \frac{1}{2} \times (60 - 40) \times V = 1000$

1 mark
Rectangle plus triangle.

$40V + 10V = 1000$

$50V = 1000$

$V = 20$ m/s

1 mark
20 m/s.

(Total for Question 18 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
EA